Xiaohan YAN

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Positions

Institut de Mathématiques de Toulouse — Toulouse, France
Postdoctoral Researcher (CNRS, mentor: Adrien Sauvaget) — *September 2025 -*

Institut de Mathématiques de Jussieu-Paris Rive Gauche — Paris, France
Postdoctoral Researcher (ERC, mentor: Penka Georgieva) — *August 2022 - August 2025*

Education

University of California, Berkeley — Berkeley, CA, USA
Ph.D. in mathematics (Advisor: Alexander Givental) — *July 2017 - May 2022*

Peking University — Beijing, China
B.Sc. in mathematics (Advisor: Huijun Fan) — *August 2013 - May 2017*

Publications

Mathematics

- **Elliptic Gromov-Witten invariants of Hilbert scheme of points on $\mathcal{A}_n$-surface** (with Hsian-Hua Tseng). Forthcoming.
- **Enumerative geometry of real r-spin curves and Landau-Ginzburg/Calabi-Yau correspondence** (with Penka Georgieva). Forthcoming.
- **Quasimap K-theoretic quantum Serre duality and wall-crossing** (with Mark Shoemaker). Forthcoming.
- **Chow ring of the stack of plane nodal curves** (with Alessio Cela and Ajith Urundolil Kumaran). *International Mathematics Research Notices*, (6), 2025. arXiv:2405.11943 doi:10.1093/imrn/rnaf048
- **Loop space formalism and K-theoretic quantum Serre duality**. Submitted. arXiv:2401.03054
- **Quantum K-theory of flag varieties via non-abelian localization**. Submitted. arXiv:2106.06281
- **Quantum K-theory of grassmannians and non-abelian localization** (with Alexander Givental). *Symmetry, Integrability and Geometry: Methods and Applications (SIGMA)*, 17:Paper No. 018, 24 pages, 2021. arXiv:2008.08182 doi:10.3842/SIGMA.2021.018
- **Relative Morse categorification theory** (with Danning Lu). Undergraduate research. arXiv:1611.06471

Physics

- **A correspondence between the quantum K theory and quantum cohomology of Grassmannians** (with Wei Gu, Jirui Guo, Leonardo Mihalcea, and Yaoxiong Wen). *Journal of Geometry and Physics*, 210:105437, 2025. arXiv:2406.13739 doi:10.1016/j.geomphys.2025.105437

Other

- **Large-Scale Automatic K-Means Clustering for Heterogeneous Many-Core Supercomputer** (with Teng Yu, Wenlai Zhao, Pan Liu, Vladimir Janjic, Shicai Wang, Haohuan Fu, Guangwen Yang and John Thomson). *IEEE Transactions on Parallel and Distributed Systems*, 31(5):997–1008, 2020. doi:10.1109/TPDS.2019.2955467

Invited Talks

Caltech/USC Algebra & Geometry Seminar
California Institute of Technology *November 2024*

Geometry Seminar
Max Planck Institute for Mathematics in the Sciences *November 2024*

Algebraic Geometry and Number Theory Seminar
Institute of Science and Technology Austria *October 2024*

Workshop "Symplectic, real, and tropical aspects of enumerative geometry"
Paris *September 2023*

Symplectix
Paris *March 2023*

Séminaire de Géométrie Énumérative
Institut de Mathématiques de Jussieu-Paris Rive Gauche *September 2022*

Mathematical Physics Seminar
Beijing International Center for Mathematical Research *August 2022*

Topology Seminar
Texas A&M University *February 2022*

Enumerative Geometry Seminar
Columbia University *January 2022*

Algebra Seminar
University of Oregon *November 2021*

Geometry Seminar
Korea Institute for Advanced Study *September 2021*

Physically Inspired Seminar
University of North Carolina, Chapel Hill *September 2021*

Geometry and Physics Seminar
Institute for Advanced Study in Mathematics, Hangzhou *June 2021*

Teaching

University of California, Berkeley

Math 54: Linear algebra and differential equations
Graduate Student Instructor *Spring 2022*

Math 54: Linear algebra and differential equations
Graduate Student Instructor *Spring 2021*

Math 54: Linear algebra and differential equations
Graduate Student Instructor *Spring 2020*

Math 54: Linear algebra and differential equations Outstanding GSI Award
Graduate Student Instructor *Fall 2019*

Language

Mandarin *(native)*, English *(fluent)*, French *(fluent)*, Latin *(written)*.

Skills

Python, C, Mathematica, LaTeX.